

MIDI Composer

User Manual

Easy Midi Composer – compose with chords, scales and built-in instruments

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1. Welcome

MIDI Composer is a lightweight program for writing music with MIDI – ideas, chord progressions and complete arrangements – without having to set up a big, complicated DAW project. You can record from a MIDI keyboard, draw notes with the mouse, let the program build chords from a scale, and hear everything with the built-in instruments or your own plugins.

When you are done, you can export a MIDI file that every music program can open, or MusicXML to view and print the score in, for example, MuseScore.

What it can do

- Multiple tracks, each with its own instrument, colour, volume, pan and MIDI channel.
- A piano roll for drawing, moving, lengthening, copying and deleting notes.
- **Chord mode**: one key plays a whole chord in the selected scale – with inversions, 7ths and 9ths.
- Next-chord suggestions based on the circle of fifths.
- Recording from a MIDI keyboard, also in a loop.
- 22 built-in band instruments (Rock, Pop, Jazz) – plus your own SoundFont files (.sf2).
- Plugins: VST3, Audio Unit (Mac), LV2 (Linux) and CLAP.
- Import and export of MIDI and MusicXML (MuseScore, Sibelius, Finale, Dorico).
- Undo/redo for everything.

Buttons and menus are written in **bold** in this manual, exactly as they appear in the program. A Danish version of this manual can be downloaded from the project's Releases page on GitHub.

2. Installation

Mac

1. Unzip **MIDI-Composer-macOS-Universal.zip**. It runs on both Intel and Apple Silicon Macs.
2. Move **MIDI Composer.app** to your **Applications** folder.
3. The program is not signed by Apple, so macOS blocks it the first time. Open Terminal and run:

```
xattr -dr com.apple.quarantine "/Applications/MIDI Composer.app"
```

If macOS still says the program cannot be opened, or that it "may harm your computer", allow it in System Settings:

1. Try to open **MIDI Composer** once and close the warning (choose **OK** or **Done**, not **Move to Trash**).
2. Open **System Settings** → **Privacy & Security**.
3. Scroll down to the **Security** section. It says that "MIDI Composer" was blocked. Click **Open Anyway**.
4. Confirm with your password or Touch ID, then click **Open** in the next dialog.

You only need to do this once per version of the program. The **Open Anyway** button is only shown for about an hour after you tried to open the program – if you cannot see it, try opening the program again first.

> **Tip:** On older macOS versions (14 and earlier) you can instead right-click the program, choose **Open** and confirm. From macOS 15 on this no longer works – use System Settings as described above.

Windows

Unzip **MIDI-Composer-Windows-x64.zip** into a folder and start **MIDI Composer.exe**. The file **GeneralUser-GS.sf2** must stay in the same folder as the program – it contains the built-in instruments.

Debian and Ubuntu

```
sudo apt install ./midi-composer__amd64.deb
```

The program then appears in the application menu as **MIDI Composer**, or can be started with the command **midi-composer**.

3. First start

1. The first time, the program asks whether to **scan for instrument plugins**. If you have plugins installed (e.g. Kontakt), choose **Scan**. You can always do it later via **Options** → **Rescan plugins**.
2. Open **Options** → **Settings (audio, MIDI, modifier keys)** and choose your audio device and MIDI keyboard (see chapter 13).
3. A new project has one track with **Grand Piano**. Play your keyboard, or click in the piano roll to add notes.

> **Tip:** No sound? Check that the right audio device is selected in Settings, that the track is not muted (M), and that the track volume is not all the way down.

4. The window

The window has these parts, from top to bottom:

Part	What it does
Menu bar	**File** (projects, import/export), **Edit** (undo, copy, tracks) and **Options** (settings, plugins, sounds, manual, about). On a Mac the menu is at the top of the screen.
Transport bar	Play, stop, record, loop, metronome, count-in, tempo, time signature and undo.
Chord panel	Root, scale, chord mode, modifier buttons, the current chord and next-chord suggestions.
Track list (left)	All tracks with name, colour, mute/solo, volume, pan, channel and instrument.
Editor bar	Above the piano roll: the **Zoom** buttons (chapter 8), **Grid** and **Snap**.
Piano roll (right)	The notes of the active track, with the timeline on top, the keyboard on the left and the velocity lane at the bottom.
Step sequencers (bottom)	Every step sequencer track, below the piano roll and always visible: the setup card on the left, the step rows aligned with the piano roll grid (chapter 9).

5. Transport, tempo and loop

Button / field	Function
|< (Return to start)	Jumps to the start – or to the loop start if loop is on. The piano roll follows.
Play	Starts playback from the playhead.
Stop	Stops. Press **Stop** again while stopped to jump back to the start.
Rec	Starts recording on the active track (see chapter 11).
Loop	Turns the loop on/off. Start and end bar are set in the **Loop bars** fields.
Click	Metronome on/off (key **K**). The first beat of each bar is accented. **Right-click** for the volume (Low, Medium, High, Max).
Count-in	Bars of metronome count-in before recording starts: none, 1, 2 or 4 bars.
Position display	Shows bar, beat and time of the playhead.
BPM	Tempo in beats per minute.
Sig	Time signature, e.g. 4/4, 3/4 or 6/8.
Undo / Redo	Undo and redo.

The timeline

- **Click** in the timeline above the piano roll to move the playhead.
- **Shift+drag** or **right-drag** in the timeline to set a loop range.

6. Tracks

Adding and deleting

Click **+ Track** and choose **Piano roll** or **Step sequencer** (also in the Edit menu). **- Track** or **Edit → Delete active track** deletes the active piano roll track. Click a track in the track list to make it active – that is the track you see and edit in the piano roll, and the one your MIDI keyboard plays.

Track settings

Element	Function
Colour button	The small square next to the name. Opens the colour picker (see below).
Name	Double-click and type to rename the track.
M / **S**	Mute (silence the track) and Solo (hear only this track).
Channel	MIDI channel 1–16. Used for export and by plugins that listen on specific channels.
Volume / Pan	The sliders for volume and panning (left/right).
Instrument button	Shows the instrument's name. Click it to choose an instrument (see chapter 7).
E (editor)	Opens the plugin's own window, if it has one (e.g. Kontakt).

Colours

Click a track's colour button to open the **colour picker**:

- Drag in the colour field and sliders, or type a hex value. The track changes colour immediately.
- **Presets**: ten fixed colours.

- **Saved**: your own swatches. Click **Save swatch** to store the current colour. They are kept in your settings and can be used in every project. Right-click a swatch to remove it.
- **Default** resets to the default colour.
- Click outside the box to close it. The colour change can be undone in one step.

The active track is shown in a strong colour – both in the track list and in the piano roll. The other tracks are muted, and their notes shine faintly through the piano roll in their own colour, like looking through tracing paper. You can see them, but not edit them.

7. Instruments

Click a track's instrument button. The menu has these groups:

Menu	Contents
Internal → Rock / Pop / Jazz	The built-in band instruments (table below).
Internal → Basic Synth	A simple, lightweight synth without samples.
SoundFonts	Your own .sf2 files – one sub-menu per file – and **Open Sounds folder**.
VST3, AudioUnit, LV2, CLAP	Your installed plugins, once scanned.
Browse plugins...	A window with search and filtering by format.
Bypass / **Remove instrument**	Temporarily disable the instrument, or remove it.

Built-in instruments

Rock	Pop	Jazz
Rock Drums	Pop Drums	Jazz Drums
Distortion Guitar	Grand Piano	Brush Drums
Overdrive Guitar	Electric Piano	Jazz Piano
Clean Electric Guitar	Acoustic Guitar	Upright Bass
Electric Bass (Pick)	Ukulele	Jazz Guitar
Rock Organ	Electric Bass (Finger)	Tenor Sax
	Strings	Trumpet
		Vibraphone
		Violin

Drum kits

The drum kits follow the General MIDI standard, so drum tracks from other programs fit directly. The most important keys:

Key	Sound	Key	Sound
C1	Bass drum	F♯1	Closed hi-hat
C♯1	Side stick	G♯1	Pedal hi-hat
D1	Snare	A♯1	Open hi-hat
D♯1	Hand clap	C♯2	Crash cymbal

Key	Sound	Key	Sound
E1	Snare 2	D♯2	Ride cymbal
F1, G1, A1, B1, C2, D2	Toms (low → high)	F2	Ride bell

Tip: Turn chord mode off on drum tracks – otherwise one key becomes three drums. With the default modifier keys C1/D1 (see chapter 10), those keys are exactly the bass drum and snare.

Your own sounds (SoundFonts)

1. Choose **Options** → **Open Sounds folder**. The folder opens in Finder/Explorer.
2. Copy one or more **.sf2** files into the folder. Many free SoundFonts are available online.
3. Choose **Options** → **Rescan plugins**.
4. The sounds now appear under **SoundFonts** in the instrument menu, sorted by file name.

Projects remember which file and sound each track uses. If you move or delete the file, the track is shown as missing until the file is back.

Plugins

The program can use VST3 plugins on all systems, Audio Units on Mac, LV2 on Linux, and CLAP. After installing new plugins: **Options** → **Rescan plugins**. If a plugin crashes during the scan, it is automatically blacklisted and skipped next time.

If a plugin used by a project is missing, the track shows **Plugin missing**. The notes are still there – choose another instrument, or reinstall the plugin.

8. Piano roll

Notes

Action	How
Add a note	**Click** an empty spot. Keep the button down and drag right to set the length. In chord mode a whole chord is inserted.
Move	Drag a note. Up/down changes pitch, sideways changes time.
Change length	Drag the note's **right edge** to change where it ends, or its **left edge** to change where it starts. Over an edge the mouse pointer changes to a ↔ resize arrow.
Select	Click a note. **Shift+click** adds to/removes from the selection.
Select several	⌘**/Ctrl+drag** or **Shift+drag** on an empty spot draws a selection box. ⌘**/Ctrl+A** selects everything.
Delete one note	**Right-click** the note.
Delete many	Hold the **right mouse button** and drag a **red box** over the notes. Every note the box touches is deleted when you release. Or select them and press **Delete**/**Backspace**.
Transpose	↑ **/** ↓ moves selected notes by a semitone. **Shift+**↑ **/** ↓ moves them an octave.
Copy / cut / paste	⌘**/Ctrl+C**, ⌘**/Ctrl+X**, ⌘**/Ctrl+V**. Pastes at the playhead.

On a Mac trackpad, a right-click is a two-finger click or Ctrl+click.

Grid and snap

Grid and **Snap** are in the bar above the piano roll. Grid: 1/4, 1/8, 1/16, 1/32 and triplets (1/8 T, 1/16 T). When **Snap** is on, new and moved notes snap to the grid.

Velocity

The lane at the bottom shows each note's velocity as a vertical line. Click and drag across the lines to change it. If some notes are selected, only the selected notes change. Soft notes are drawn slightly darker in the piano roll.

Zoom and scroll

Mouse / trackpad	Effect
Scroll	Scroll up/down (and sideways with a trackpad).
Shift+scroll	Scroll horizontally in time.
⌘/Ctrl+scroll	Zoom in/out in time.
Alt/⌥+scroll	Zoom vertically (taller/shorter rows).
Scroll in the timeline	Zoom in time.
Zoom out / Zoom in	Toolbar buttons: zoom in time. The **+** and **-** keys do the same.
Lower / Taller	Toolbar buttons: smaller or taller note rows. The keyboard follows the rows.

The keyboard

Click the keys on the left to hear the notes. The notes of the selected scale are highlighted as lighter rows in the piano roll, so you can easily see which notes fit.

9. Step sequencer

The step sequencer is a quick way to build drum beats and other repeating patterns. A step sequencer track is built from **lines**. Each line plays one note with its own instrument, e.g. kick, snare and hi-hat, or a bass note on a bass instrument.

All step sequencers are shown **below the piano roll** and are always visible. Their step rows follow the piano roll grid – same bars, scroll and zoom – so you can see exactly where the rhythm falls under your notes. The piano roll keeps its size: the first step sequencer is shown just below it, and further step sequencers follow underneath – use the scrollbar on the far right of the window to scroll down to them.

Getting started

1. Click **+ Track** in the track list and choose **Step sequencer** (or **Edit** → **Add step sequencer track**). It appears below the piano roll with a **Kick** line on Pop Drums.
2. Click **+ Line** on its card to add more lines. New lines suggest Snare, Closed Hat, Open Hat, Clap and so on, and use the same instrument as the line above.

3. Click the steps to switch them on. Press **Space** to play – the red playhead runs through the piano roll and the step rows together.

The card

The setup of each step sequencer is on its card in the left column:

Setting	Effect
Title	Double-click to rename the step sequencer, e.g. **Drums verse**.
▾ / ▸	Collapses the step sequencer to its top row (or expands it again).
Colour dot	Click to choose the colour of the step sequencer – handy when you have several.
+ Line	Adds a line.
X (top)	Deletes the whole step sequencer with all lines (can be undone).
Step	Length of each step: 1/4, 1/8, 1/16, 1/32 or triplets. With 1/16 a 4/4 bar has 16 steps. Changing it keeps your rhythm where possible.
Bars	Length of the step sequencer in bars. **Every bar has its own steps** – change one bar without affecting the others. New bars start empty.
Start	The bar of the song where the step sequencer starts.

Working with bars

The strip above the step rows shows each bar (**Bar 1**, **Bar 2** ...). **Right-click** a bar there for these commands (they apply to all lines):

Command	Effect
Copy to next bar	Copies the bar onto the following bar – a quick start for a variation.
Duplicate bar	Inserts a copy of the bar right after it.
Add empty bar after	Inserts an empty bar after it.
Clear bar	Switches off all steps in the bar.
Delete bar	Removes the bar; the following bars move left.

A line

Part	What it does
Name	Double-click to rename the line.
Instrument button	Choose the line's instrument – the same menu as in the track list (chapter 7).
M / **X**	Mute the line / delete the line (can be undone).
Note	Under the piano keyboard: the note the line plays. For drum kits: C2 kick, D2 snare, F♯2 closed hi-hat, A♯2 open hi-hat.
Steps	**Click** = on/off. **Shift+click** = accent (louder). **Right-click** = off. **Drag** across several steps to switch them all on or off.

Tip: The notes are generated from the pattern, so they play and export to MIDI like any other notes. ⌘**/Ctrl+scroll** over the step rows zooms the time axis, **Shift+scroll** scrolls it – the piano roll follows.

10. Chord mode

In chord mode, every key – on the MIDI keyboard or in the piano roll – plays a whole chord that fits the selected scale. Play C in C major and you get a C major chord. Play D and you get D minor, and so on.

Turning it on

1. Choose the root in **Root** and the scale in **Scale** (e.g. A Natural Minor).
2. Click **Chord Mode** so it shows ON – or press **C** on the computer keyboard.
3. Play or click a note in the scale. Notes outside the scale are played as single notes.

Scales: Major, Natural Minor, Harmonic Minor, Melodic Minor, Dorian, Phrygian, Lydian, Mixolydian and Locrian.

Modifiers: inversions, 7ths and 9ths

The modifier buttons in the chord panel change the chord. You can click the buttons (they stay on), or hold the corresponding keys on your keyboard:

Modifier	Default key	Effect
1st inversion	C1	The top note moves down below the note you play.
2nd inversion	D1	The two top notes move down below the note you play.
7	C♯1	Adds the seventh (e.g. Am → Am7, G → G7).
9	D♯1	Adds the seventh and ninth (e.g. Am → Am9).

The note you play always stays where you put it. Example with C major played from C3:

	Notes
Root position	C3 – E3 – G3
1st inversion	G2 – C3 – E3
2nd inversion	E2 – G2 – C3

The modifier keys do not make a sound themselves in chord mode. You can choose other keys in Settings (chapter 13). The current chord is shown in the chord panel, e.g. **Am7 – 1st inversion**.

Next-chord suggestions

After you play a chord, the chord panel shows up to four suggestions next to **Next**. Click a suggestion to hear it. Hover over it to see why it is suggested. The suggestions follow the circle of fifths within the scale, strongest first:

Priority	Root movement	Example in C major
1	Down a fifth (strongest resolution)	G → C, Dm → G
2	Up a fifth	C → G
3	Down a third (two common notes)	C → Am
4	Back to the tonic	→ C

Priority	Root movement	Example in C major
5	One step up	C → Dm

11. Recording

1. Select the track you want to record on.
2. Choose a **Count-in** (e.g. 1 bar) and turn on **Click** if you want the metronome while you record.
3. Press **Rec** (or **R**). With a count-in the metronome counts the bars first – the display shows **COUNT-IN** and the beats left – then playback and recording start. What you play is shown in red.
4. Press **Stop** (or **space**). The notes are added to the track in one step, which can be undone.

If **Loop** is on, recording keeps going around the loop, and everything you play on each pass is kept. That way you can build up layers, e.g. first the bass and then the chords. Chord mode and modifiers also work while recording.

> **Tip:** Notes played during the count-in are ignored, except notes played just before the first beat – they are recorded at the start, so a slightly early first note is not lost.

12. Files, import and export

Menu	Function
File → New / Open / Save / Save as	Projects are saved as **.mcproj**. Tracks, notes, instruments with their settings, colours, tempo and loop are included.
File → Import MIDI file...	Adds the tracks of a .mid file to the project.
File → Export song... (⌘/Ctrl+E)	Saves the finished song as WAV, MP3, OGG, FLAC, MIDI or M4A (see below).
File → Export MIDI file...	Saves all tracks as a standard MIDI file that any music program can open.
File → Import MusicXML (MuseScore)...	Loads .musicxml, .xml or .mxl. Each part becomes a track.
File → Export MusicXML (MuseScore)...	Saves a score that opens in MuseScore, Sibelius, Finale and Dorico.

Exporting the finished song

File → Export song... turns the song into an audio file you can play anywhere, share or put on a CD. The song is rendered faster than real time through all instruments, effects and the mixer, so the file sounds exactly like playback: mute, solo, volume, pan, EQ, compressor, reverb, volume lines and audio clips are included. A progress window shows how far it is; **Cancel** stops it.

Format	Use
WAV	Uncompressed, the best quality – for CDs, video and further work. 16, 24 or 32-bit float.
FLAC	Lossless: the same sound as WAV in about half the size.

Format	Use
MP3	Plays everywhere; small files. 320 kbps sounds practically like the original.
OGG Vorbis	Free, open format with small files and very good sound.
M4A (AAC)	As used by Apple Music and iTunes; small files with very good sound.
MIDI	Notes only, no sound – for other music and notation programs.

- **Range**: the whole song, or only the part between the loop markers.
- **Let reverb and released notes ring out**: keeps rendering after the last note until the sound has died away.
- **Normalize**: makes the song as loud as possible without distortion (loudest peak at -1 dB). If the song is too loud and would distort, the program tells you after the export.
- WAV, FLAC and OGG need nothing extra. M4A uses the encoder built into macOS. MP3 – and M4A on Windows and Linux – needs the free program **LAME** or **FFmpeg**. It is found automatically; otherwise the program shows how to install it, or you choose it with **Locate encoder...**.

MusicXML – to and from sheet music

Export: the key (from Root/Scale), time signature and tempo are included. Notes are rounded to the nearest 1/16, notes across bar lines are tied, and chords are written as chords. Piano tracks with a wide range get a grand staff.

Import: notes, chords, rests, ties, pickup bars and multiple voices are included, and the program sets Root/Scale from the score's key signature. Repeats are not unrolled, and grace notes and dynamics are skipped.

In MuseScore, use **File** → **Export** → **MusicXML** (not the normal .mscz file) to bring the score into MIDI Composer.

13. Settings

Options → **Settings (audio, MIDI, modifier keys)** contains:

- **Audio device**: output, sample rate and **buffer size**. A small buffer (128–256 samples) gives a short delay when you play. If you hear crackles, choose a larger one.
- **MIDI input**: your MIDI keyboard. The program reconnects automatically if the keyboard is unplugged and plugged in again.
- **Chord modifier keys**: which keys act as 1st inversion, 2nd inversion, 7 and 9. Click **MIDI Learn** and press the key you want to use. **Defaults** restores the default keys C1, D1, C♯1 and D♯1.

Options → **User manual** opens this manual. **Options** → **About MIDI Composer** shows the version and credits.

14. Keyboard shortcuts

Key	Function
Space	Play / stop
Home	To start (or loop start)

Key	Function
R	Record on/off
L	Loop on/off
K	Metronome on/off
C	Chord mode on/off
+ / -	Zoom in / out in time (piano roll)
⌘/Ctrl+Z	Undo
⌘/Ctrl+Shift+Z or ⌘/Ctrl+Y	Redo
⌘/Ctrl+N / O / S	New project / Open / Save
⌘/Ctrl+Shift+S	Save as
⌘/Ctrl+E	Export song (WAV, MP3, OGG, FLAC, MIDI, M4A)
⌘/Ctrl+A	Select all notes
⌘/Ctrl+C / X / V	Copy / cut / paste notes
Delete / Backspace	Delete selected notes
↑ / ↓	Transpose selected notes by a semitone
Shift+↑ / ↓	Transpose selected notes by an octave

⌘ applies on Mac, Ctrl on Windows and Linux.

15. Troubleshooting

Problem	Solution
No sound	Check the audio device in Settings, mute/solo and volume on the track, and that the track has an instrument.
The sound comes late	Choose a smaller buffer size in Settings (e.g. 256 or 128 samples).
The sound crackles	Choose a larger buffer size, or use fewer heavy plugins at once.
The MIDI keyboard does nothing	Select it under MIDI input in Settings. Check that the right track is active.
One key plays three notes	Chord mode is on – press **C** or click **Chord Mode**.
A key plays no sound	It is set as a modifier key in chord mode. Turn chord mode off, or choose other modifier keys in Settings.
A plugin is missing from the list	Run **Options** → **Rescan plugins**. Plugins that crashed during the scan are skipped.
My own .sf2 sounds do not appear	The file must be in the Sounds folder and end in .sf2. Then run **Rescan plugins**.
Mac: "cannot be opened"	Run the command from chapter 2 (xattr), or go to **System Settings** → **Privacy & Security** and click **Open Anyway** (see chapter 2).
The built-in instruments are missing	The file GeneralUser-GS.sf2 must be next to the program (Windows/Linux) or inside the app (Mac). Reinstall the program.

16. Credits and licenses

- **GeneralUser GS** – the built-in instruments – by S. Christian Collins. https://www.schristiancollins.com/generaluser. The license is included as GeneralUser-GS-LICENSE.txt.
- **Valiant Violin v2 GM** – the built-in Violin (Jazz) – by Giovanni Valentino with Mike77154 (WTFPL license). https://musical-artifacts.com/artifacts/1796. The license is included as Valiant-Violin-LICENSE.txt.
- **TinySoundFont** – SoundFont playback – by Bernhard Schelling (MIT license). https://github.com/schellingb/TinySoundFont
- **JUCE** – the framework MIDI Composer is built with. https://juce.com
- Source code: https://github.com/shenphenchoedron-hue/easymidieditor